

# Breaking the computational barrier in high-resolution weather forecasting with decoupled training

Kang Chen<sup>a,b</sup>, Fenghua Ling<sup>b,\*</sup>, Ben Fei<sup>b,c,\*</sup>, Hao Chen<sup>b</sup>, Shaochen Wang<sup>d</sup>, Lei Liu<sup>a</sup>, Bin Li<sup>a</sup>, Lei Bai<sup>b</sup>

<sup>a</sup> University of Science and Technology of China, Hefei, China

<sup>b</sup> Shanghai Artificial Intelligence Laboratory, Shanghai, China

<sup>c</sup> The Chinese University of Hong Kong, Hong Kong, China

<sup>d</sup> Jiangxi Normal University, Nanchang, China

## ARTICLE INFO

### Keywords:

Weather prediction  
High resolution  
Tropical cyclone  
Machine learning

### 关键词:

天气预报  
高分辨率  
热带气旋  
机器学习

## ABSTRACT

Achieving high-resolution global AI weather forecasts remains computationally prohibitive, constrained by extensive GPU memory demands and prolonged training times. Existing approaches impose trade-offs—balancing speed, memory, or numerical precision. To circumvent these trade-offs, the authors introduce DeTI, a Decoupled Training–Inference framework that trains exclusively on high-resolution regional patches while incorporating global context from a pretrained, frozen low-resolution model. DeTI reduces computational costs yet preserves fidelity. Remarkably, it generalizes seamlessly during inference, enabling the extension of regional learning to global forecasts. A 0.25° model trained in only two days on eight NVIDIA A100 GPUs attains accuracy comparable to state-of-the-art systems. Moreover, DeTI scales efficiently to 0.09° resolution, demonstrating superior capability in high-impact scenarios, excelling in predicting typhoon intensity and capturing the fine-grained structure of heatwaves, paving the way for future kilometer-scale AI weather models.

### 摘要

针对全球高分辨率AI天气预报显存消耗大, 训练周期长等瓶颈, 本文提出DeTI (解耦训练-推理) 框架。该框架打破了训练与推理分辨率必须一致的限制, 采用“区域高分辨率训练”辅以“全球低分辨率背景引导”策略, 实现了计算开销的有效解耦。实验证实, 仅需8块A100 GPU训练2天, 0.25°模型精度即可媲美主流大模型。此外, DeTI可扩展至0.09°分辨率, 在台风强度预测及热浪精细表征中表现优异, 为开发公里级AI天气模型提供了高效路径。

## 1. Introduction

Accurate high-resolution weather forecasting is essential for societal resilience and economic planning (Esteves et al., 2023; Rasp et al., 2020), providing actionable insights for agriculture, logistics, and disaster risk reduction (Alley et al., 2019). In recent years, data-driven methods, particularly deep learning (Dosovitskiy et al., 2020; He et al., 2016; Schmidt, 2019), have revolutionized weather forecasting. Pioneering models such as Pangu-Weather (Bi et al., 2023), GraphCast (Lam et al., 2022), and GenCast (Price et al., 2023) have demonstrated remarkable success, achieving forecast accuracy comparable to, and even surpassing, traditional NWP (numerical weather prediction) models (Charney et al., 1950) while requiring a fraction of the computational cost during inference.

However, this success brings a fundamental challenge that limits its future progress. The main barrier is the prohibitive computational

cost of model training (Ben-Bouallegue et al., 2023). Building large-scale models requires intensive gradient computations and the storage of intermediate results. As a result, current state-of-the-art methods depend on specialized strategies that save computational resources. These approaches involve difficult trade-offs (Micikevicius et al., 2018; Zhao et al., 2023): some reduce numerical precision to lower memory consumption, while others share model components across multiple devices to save memory, which increases communication and slows the overall process (Bodnar et al., 2024; Chen et al., 2023b; Kochkov et al., 2024; Ramavajjala, 2024; Zhong et al., 2024). While these methods mitigate GPU memory constraints, they will further extend the training process. For example, Pangu-Weather needs 192 V100 GPUs for nearly one month of training. GraphCast requires four weeks on 32 TPUv4 devices. Such extreme computational demands make it difficult to develop the next generation of high-resolution models. Here, we address this computational barrier with the Decoupled Training–Inference (DeTI) frame-

Peer review under the responsibility of Editorial Board of Atmospheric and Oceanic Science Letters.

\* Corresponding authors.

E-mail addresses: [lingfenghua@pjlab.org.cn](mailto:lingfenghua@pjlab.org.cn) (F. Ling), [benfei@cuhk.edu.hk](mailto:benfei@cuhk.edu.hk) (B. Fei).

<https://doi.org/10.1016/j.aosl.2026.100821>

Received 3 December 2025; Revised 3 February 2026; Accepted 16 March 2026

Available online xxx

1674-2834/© 2026 The Authors. Publishing Services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Please cite this article as: K. Chen, F. Ling, B. Fei et al., Breaking the computational barrier in high-resolution weather forecasting with decoupled training, Atmospheric and Oceanic Science Letters, <https://doi.org/10.1016/j.aosl.2026.100821>

work, an efficient and scalable paradigm that avoids directly training on full high-resolution global data. Unlike efficiency-oriented strategies such as knowledge distillation (Hinton et al., 2015) or dynamic resolution training (Bodnar et al., 2024), DeTI offers a distinct, orthogonal approach focused on decoupling training data and gradient flow. DeTI's main innovation is to train only a high-resolution module on small, regional "patches" to learn fine-grained physical details. This module is guided by a pretrained, frozen low-resolution global model that provides the necessary global context. By feeding intermediate features from the frozen backbone to the high-resolution module, this approach substantially reduces GPU memory and computational costs compared to standard full-grid training. This efficiency is primarily driven by training on smaller regional patches rather than the entire high-resolution domain, combined with the fact that gradients are computed only for the trainable module while the global backbone remains frozen. During inference, the regionally trained high-resolution module generalizes, seamlessly generating complete, high-resolution global forecasts.

Our results show that DeTI achieves a strong balance of efficiency and accuracy. The framework trains a  $0.25^\circ$  model in just two days on eight NVIDIA A100 GPUs, using only a small fraction of the resources required by current state-of-the-art models. At the same time, DeTI maintains forecast accuracy comparable to models such as Pangu-Weather and GraphCast. Its full potential is evident in its scalability, demonstrated by performance on  $0.09^\circ$  resolution data that surpasses traditional physics-based models. This work provides a practical and resource-efficient path forward and makes the development of next-generation, kilometer-scale AI weather models feasible without the need for massive specialized computing clusters.

## 2. Data and methods

### 2.1. Dataset

We utilize two datasets from the European Centre for Medium-Range Weather Forecasts (ECMWF). The first is the fifth generation ECMWF atmospheric reanalysis (ERA5) (Hersbach et al., 2020) at  $0.25^\circ$  resolution ( $721 \times 1440$  grid), and the second is the high-resolution operational analysis (HRES) at  $0.09^\circ$  resolution ( $2001 \times 4000$  grid). While both datasets share a consistent 6-h sampling interval, we employ distinct data splitting protocols to accommodate their differing temporal coverages. For the standard DeTI model (ERA5), we use the years 1979–2017 for training and 2018 for testing. Conversely, for the DeTI-Large model (HRES), limited historical availability necessitates a shift to 2016–2021 for training and 2022 for testing (see Text S1). Despite these differences in resolution and time, the input feature set is unified: all models utilize four surface-level variables (U10, V10, T2m, MSL) and five upper-air variables (U, V, Z, Q, T) across 13 pressure levels. For detailed preprocessing steps, refer to Text S2.

### 2.2. Methods

Our proposed DeTI framework is a dual-branch architecture (Fig. 1) designed to strategically decouple high-resolution training from the global context. We first describe the components, then the training and inference workflows.

#### 2.2.1. Architectural components

DeTI consists of two core components: the Global-Context Low-Resolution Network (GC-LRN) and the Detail-Enhanced High-Resolution Network (DE-HRN). The GC-LRN represents the frozen context within the system. It is a pretrained global model whose parameters remain fixed during the training of DeTI. Its primary function is to process low-resolution global states and to provide large-scale atmospheric context that guides subsequent stages of learning like a teacher. The DE-HRN serves as the trainable learner. It processes small, high-resolution regional patches and learns fine-grained physical details that are conditioned on the contextual information provided by the GC-LRN.

Both networks are built upon the Swin U-Net architecture described by Ronneberger et al. (2015) and Liu et al. (2021). This architecture consists of an encoder that includes convolutional embeddings, Swin blocks, and downsampling operations, together with a symmetric decoder. Following the approach of FengWu (Chen et al., 2023a), both components employ a multi-input, multi-output configuration to enable deep feature fusion (for more details, see Text S3).

The critical link between the two networks is the Global-Context Injection (GCI) Module (Fig. 1(b)). The GCI dynamically integrates relevant low-resolution context. First, we use spatial correspondence to align the low-resolution feature map  $F_l$  with the high-resolution grid coordinates  $G_h$  via bilinear interpolation:

$$F_{l \rightarrow h} = \text{GridSample}(F_l, G_h). \quad (1)$$

This yields a feature map  $F_{l \rightarrow h}$  spatially aligned with the high-resolution feature map,  $F_h$ . Next, we employ a cross-attention mechanism (Vaswani et al., 2023) where high-resolution features act as the query (Q), while context is drawn from both high-resolution and aligned low-resolution features to form the keys (K) and values (V):

$$Q = W_Q F_h, \quad (2)$$

$$K_h = W_K F_h, \quad V_h = W_V F_h, \quad (3)$$

$$K_l = W_K F_{l \rightarrow h}, \quad V_l = W_V F_{l \rightarrow h}, \quad (4)$$

$$K_{\text{concat}} = \text{concat}(K_h, K_l), \quad V_{\text{concat}} = \text{concat}(V_h, V_l), \quad (5)$$

$$F'_h = \text{Attention}(Q, K_{\text{concat}}, V_{\text{concat}}) + F_h, \quad (6)$$

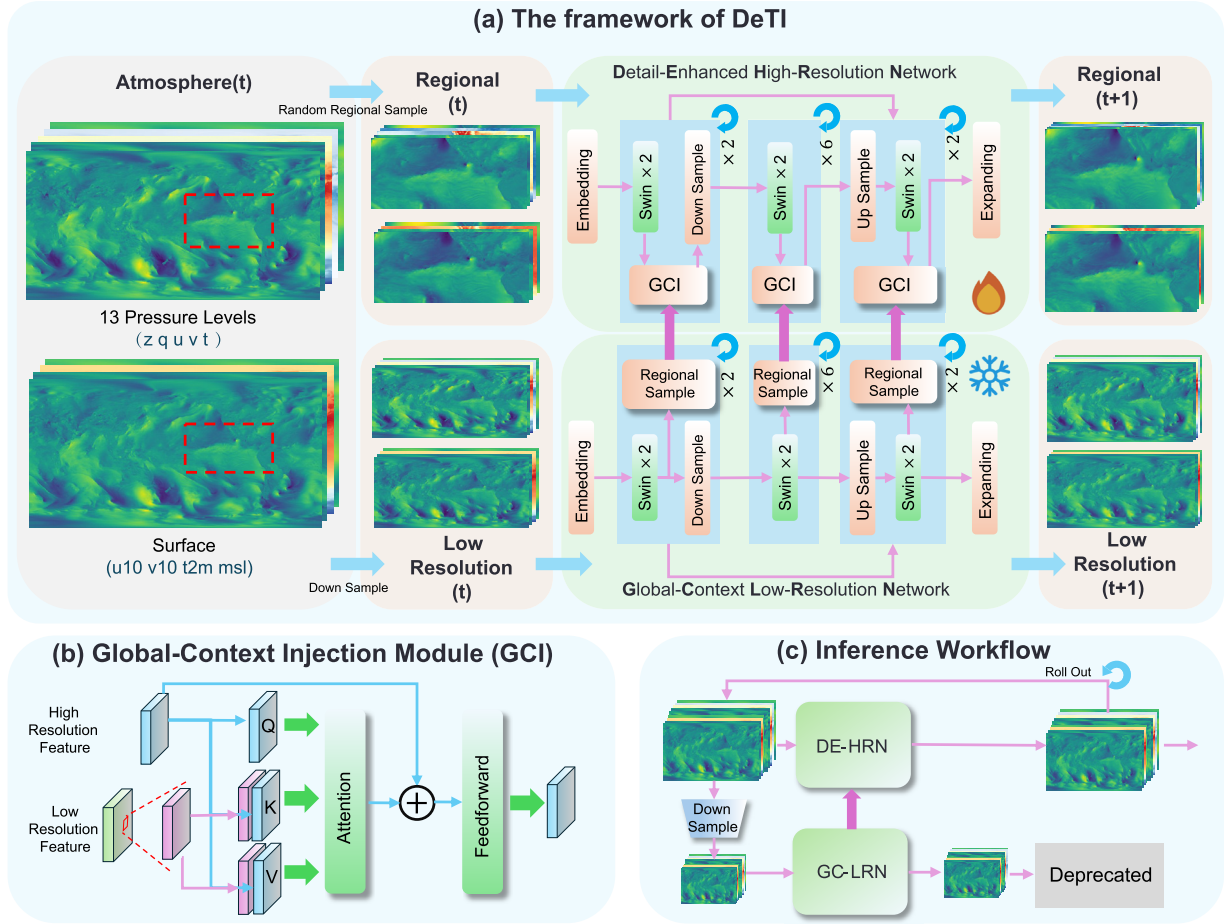
where  $W_Q, W_K, W_V$  are learnable projection matrices. The attention output is added back via a residual connection, allowing each high-resolution feature to selectively integrate local neighborhood information (from  $F_h$ ) and global context (from  $F_{l \rightarrow h}$ ) before being processed by a feedforward network.

#### 2.2.2. The DeTI workflow

During training, the task is formulated as regional high-resolution forecasting, which is augmented by context from a global low-resolution state. This strategy minimizes computational overhead by drastically reducing the spatial dimensionality of the training data. Specifically, the model takes as input a downsampled low-resolution global state  $X_{LG}^t \in \mathbb{R}^{C \times H_{LG} \times W_{LG}}$  and a randomly sampled high-resolution regional patch  $X_{HR}^t \in \mathbb{R}^{C \times H_{HR} \times W_{HR}}$  from two consecutive time steps,  $t$  and  $t-1$ . Here,  $R$  denotes the real number set,  $C$  is the number of meteorological variables, and  $H$  and  $W$  represent the latitudinal and longitudinal grid dimensions. As analyzed in Text S4, this sampling strategy incorporates dynamic longitudinal shifting to ensure uniform coverage across periodic boundaries. Furthermore, the sampling probability naturally decreases towards the poles, effectively aligning the unweighted training loss with area-weighted evaluation metrics by mitigating high-latitude over-representation. The objective is to predict the corresponding high-resolution regional patch at the next time step,  $t+1$ :

$$\hat{X}_{HR}^{t+1} = \text{DeTI}_{\text{train}}(X_{LG}^{t-1:t}, X_{HR}^{t-1:t}). \quad (7)$$

This decoupled design, enabled by the frozen GC-LRN and the unidirectional GCI module, is the key to DeTI's efficiency. Information flows from the GC-LRN to the DE-HRN without requiring backpropagation into the global context network. Gradients are computed only for the DE-HRN, leading to a training objective based solely on a high-resolution regional loss and significantly reducing GPU memory consumption. Crucially, this loss is calculated over the entire spatial extent of the patch rather than just the center region, ensuring that the model receives supervision signals across the full domain including boundaries.



**Fig. 1.** Overview of the DeTI framework. (a) The framework of DeTI: DeTI utilizes two parallel networks—the trainable Detail-Enhanced High-Resolution Network (DE-HRN) and the frozen Global-Context Low-Resolution Network (GC-LRN). During training, inputs are processed via random regional sampling and downsampling. (b) Global-Context Injection Module: The Global-Context Injection Module (GCI) injects global context into the high-resolution path using a cross-attention mechanism where queries (Q) come from DE-HRN and keys/values (K/V) come from GC-LRN. (c) Inference Workflow: During Inference, DE-HRN generates the final prediction for auto-regressive “Roll Out”, assisted by context from GC-LRN, whose own final output is deprecated.

During inference, DeTI generalizes its regionally trained knowledge to perform global high-resolution forecasting in an autoregressive manner. This generalization is structurally enabled by the design of the Swin Transformer (Liu et al., 2021). Specifically, by computing self-attention within local windows that share weights across the image, the network learns location-independent physical features rather than absolute positional mappings. Although trained on small patches, the DE-HRN can therefore be directly applied to the full high-resolution global grid without architectural modification. The validity of this patch-to-global transfer is empirically demonstrated by the consistent global forecast accuracy reported in Section 3. Leveraging this capability, the model iteratively processes the full high-resolution global state  $X_{HG}^t \in \mathbb{R}^{C \times H_{HG} \times W_{HG}}$  and its downsampled low-resolution counterpart  $X_{LG}^t$ . Starting from an initial state, the model predicts the future state  $\hat{X}_{HG}^{t+1}$ , which is then used as the input for the subsequent step:

$$\hat{X}_{HG}^{t+1} = \text{DeTI}_{\text{infer}}(X_{HG}^t, X_{HG}^{t-1}). \quad (8)$$

This iterative process produces a sequence of global high-resolution forecasts  $\{\hat{X}_{HG}^{t+1}, \dots, \hat{X}_{HG}^{t+T}\}$  over a forecast horizon of  $T$  time steps, where  $T$  denotes the total number of iterative steps in the autoregressive rollout cycle. In adherence to our design, only this high-fidelity output from the DE-HRN is forwarded to the next time step. The direct output from the low-resolution path (GC-LRN) is consequently discarded for the main autoregressive loop, ensuring the forecast chain consistently builds upon the most accurate available data (Fig. 1(c)).

### 3. Results

We evaluate the DeTI framework in terms of its central contributions—namely, forecasting accuracy, computational efficiency, and scalability to high spatial resolutions. The results are benchmarked against leading operational physics-based models and state-of-the-art AI models. To ensure a fair and direct comparison of fundamental atmospheric modeling capabilities, the primary benchmarks are based on six-hour pretrained versions of state-of-the-art models such as Pangu-Weather (6h). This design isolates the influence of architectural and training paradigms from differences in fine-tuning data. A detailed explanation of this nomenclature is provided in Texts S5 and S6.

The evaluation is performed at two key spatial resolutions. At  $0.25^\circ$  resolution, the standard DeTI model is compared with FourCastNet (Kurth et al., 2023), Pangu-Weather (6h), and GraphCast (pretrained). At  $0.09^\circ$  resolution, the DeTI-Large model is evaluated against the operational physics-based IFS-HRES system and FengWu-GHR (Han et al., 2024).

#### 3.1. Forecasting skill and training efficiency

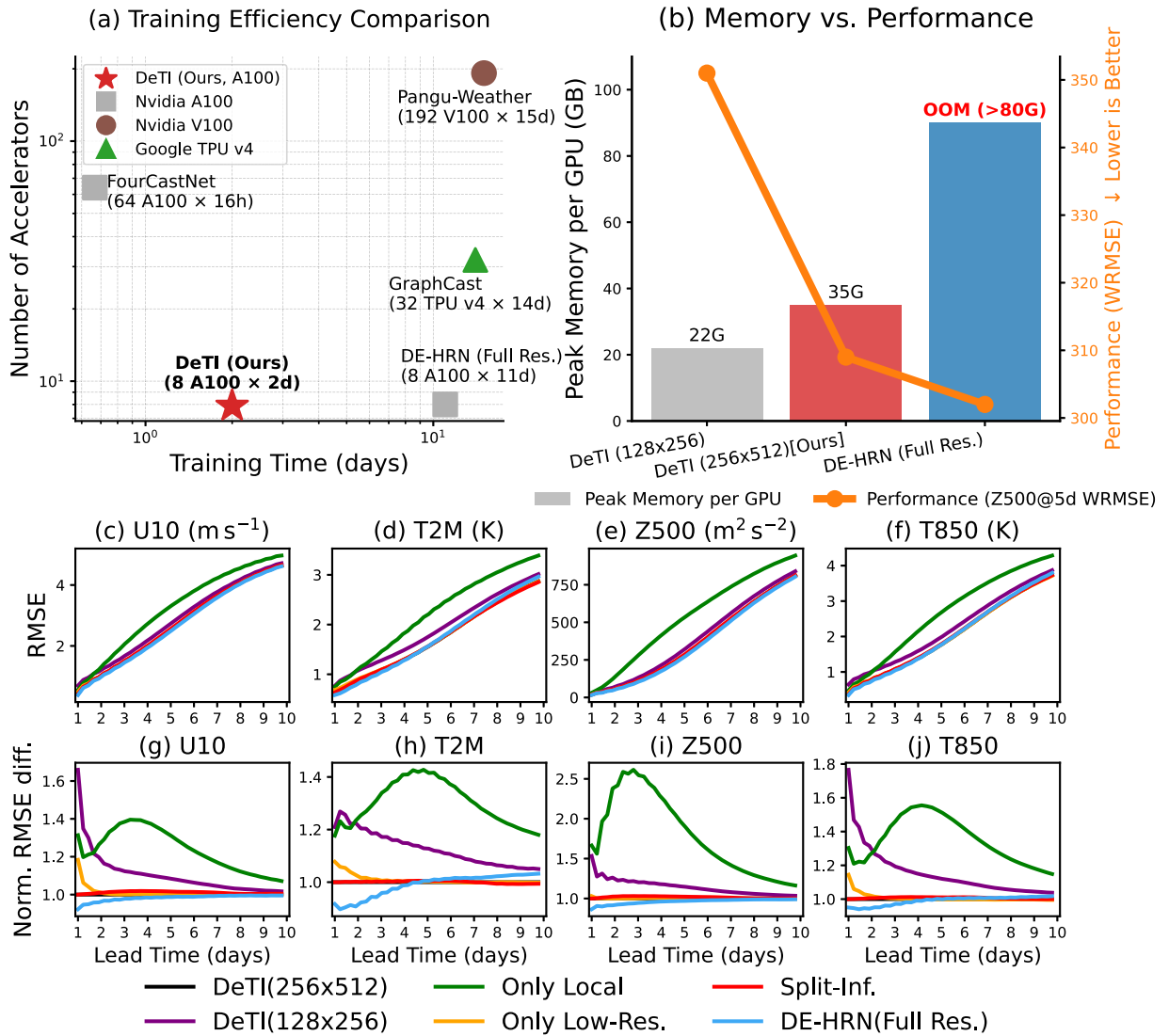
We first assess the foundational forecasting skill of the standard  $0.25^\circ$  DeTI model. The evaluation employs the weighted root-mean-square error (WRMSE) across key upper-air variables (Z500, T850) and surface variables (T2M, U10). As illustrated in Table 1, DeTI achieves forecasting accuracy comparable to, and in several cases exceeding, that of the

foundational six-hour versions of Pangu-Weather and GraphCast over three-, five-, and seven-day lead times. This performance is notable because DeTI is trained exclusively on localized, high-resolution regional patches rather than on the entire global domain. The results therefore, demonstrate that the decoupled design, which integrates global contextual information from the frozen low-resolution network with region-specific high-resolution learning, is sufficient to reproduce global high-resolution dynamics with accuracy rivaling that of full-domain state-of-the-art models. To further assess scalability, we apply the framework to a more demanding  $0.09^\circ$  resolution dataset. The results show that the DeTI-Large model maintains high predictive skill at this finer resolution. It outperforms the operational physics-based IFS-HRES system across multiple evaluation metrics and achieves accuracy comparable to the six-hourly pretrained FengWu-GHR. (Additional evaluations are presented in Texts S7–S9). It is important to note that this study focuses on assessing intrinsic model capability rather than fine-tuning optimization. Fine-tuning often demands substantial computational resources and access to large-scale operational infrastructures, which are beyond

the scope of the present work. Our aim here is to establish the scientific validity and generality of the DeTI framework as a foundational architecture. Future work may explore fine-tuning and cross-variable adaptation once sufficient resources become available.

DeTI also delivers high accuracy with improved training efficiency. As shown in Fig. 2(a), the DeTI model can be trained to convergence within only two days using eight NVIDIA A100 GPUs. This represents a marked difference compared to the resource demands of leading models such as Pangu-Weather, which requires 192 GPUs for 15 days, and GraphCast, which uses 32 GPUs for 14 days. DeTI therefore markedly lowers the computational barrier to high-resolution modeling, enabling advanced weather prediction without dependence on massive supercomputing clusters.

Finally, to better understand the sources of DeTI's efficiency and accuracy, we conducted a comprehensive ablation study to validate its core design principles (Fig. 2). Fig. 2(b) specifically quantifies the critical trade-off between memory cost and predictive skill. As shown, increasing the high-resolution patch size during training im-



**Fig. 2.** DeTI performance, training efficiency, and ablation studies. (a) Training efficiency comparison of DeTI (256×512) and DE-HRN (Full Res.) against other state-of-the-art models, plotting the number of accelerators versus total training time in days. (b) Ablation study on the training memory-performance trade-off. Bars (left axis) show the peak memory per GPU (GB) required for different high-resolution sampling strategies. The line plot (right axis) depicts the corresponding 5-day Z500 WRMSSE (weighted root-mean-square error). (c–f) WRMSSE and (g–j) normalized RMSE difference (relative to DeTI (256×512)) versus lead time (days) for key atmospheric variables: U10, T2M, Z500, and T850. Model configurations shown are: “DeTI (256×512)” (our final model, trained with  $256 \times 512$  high-resolution regional patches); “DeTI (128×256)” (trained with  $128 \times 256$  patches); “Only Local” (trained using only the high-resolution network without global context); “Only Low-Res.” (trained using only the low-resolution global model); “Split Inf.” (separate high- and low-resolution inference); and “DE-HRN (Full Res.)” (trained on global high-resolution inputs).

**Table 1** Performance comparison of our proposed DeTI against other state-of-the-art models. The entries “3/5/7” denote forecast lead times of 3, 5, and 7 days, respectively. The evaluation metric is the WRMSE (weighted root-mean-square error), where lower values indicate better performance. The best results for each variable and lead time are highlighted in bold.

| Resolution | Model                | Z500 (3/5/7)       | T850 (3/5/7)           | T2m (3/5/7)           | U10 (3/5/7)           |
|------------|----------------------|--------------------|------------------------|-----------------------|-----------------------|
| 0.25°      | FourCastNet          | 253/484/689        | 1.56/2.48/3.32         | 1.39/2.00/2.54        | 2.24/3.41/4.18        |
|            | Pangu-Weather (6h)   | 171/375/609        | 1.39/2.25/3.21         | 1.50/2.13/2.80        | 1.85/2.92/3.92        |
|            | GraphCast (pretrain) | 155/333/554        | 1.22/1.98/2.84         | 1.13/1.68/2.29        | 1.73/2.72/3.69        |
|            | DeTi (ours)          | 141/309/526        | <b>1.21/1.89/2.71</b>  | 1.15/1.60/2.16        | <b>1.69/2.66/3.63</b> |
| 0.09°      | IFS                  | 138/308/523        | 1.31/1.95/2.72         | <b>1.10/1.53/2.06</b> | 1.90/2.82/3.74        |
|            | FengWu-GHR           | 135/306/519        | 1.21/ <b>1.87/2.68</b> | 1.22/1.65/2.16        | 1.76/2.71/3.65        |
|            | DeTi-Large (ours)    | <b>133/302/517</b> | 1.24/1.91/2.72         | 1.26/1.76/2.33        | 1.77/2.73/3.68        |
|            | Climatology          | 810/810/810        | 3.41/3.41/3.41         | 2.61/2.61/2.61        | 3.91/3.91/3.91        |

proves forecast skill (lower WRMSE) but substantially increases peak GPU memory requirements. While training directly on the full global high-resolution field (DE-HRN(Full Res.)) yields the highest predictive skill (Fig. 2(b)), its prohibitive memory cost leads to OOM (out-of-memory) failures (80G) without memory-saving techniques. Moreover, even when employing such techniques (e.g., activation checkpointing), scaling full-resolution training to even higher resolutions remains computationally challenging. Further experiments (Fig. 2(c–j)) underscore the necessity of DeTI’s components. Training solely on local patches (Only Local) leads to rapid error accumulation and instability over lead time. Conversely, relying only on low-resolution inputs (Only Low-Res.) degrades initial accuracy due to missing high-frequency details. These results demonstrate that DeTI’s patch-based strategy, augmented with global context, strikes an effective balance, preserving predictive skill while managing the prohibitive computational demands of full-domain, high-resolution training (for more analysis, see Text S10).

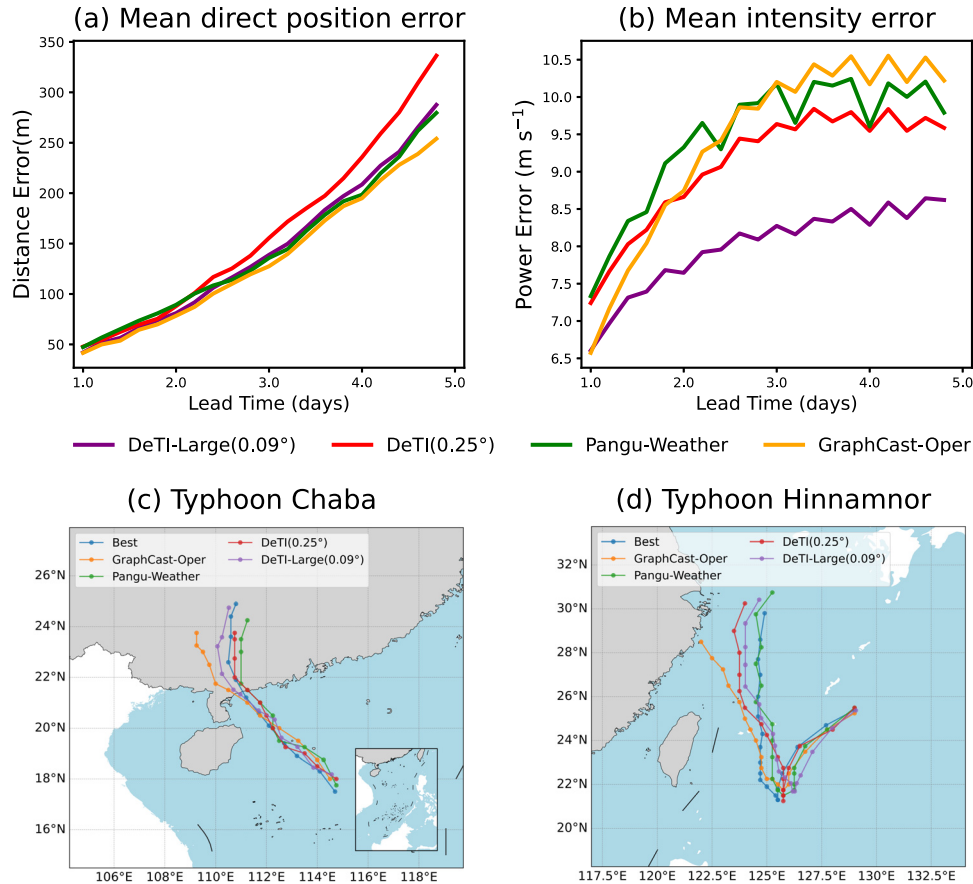
### 3.2. Forecasting high-impact extreme weather

While global metrics like WRMSE (Table 1) are essential for benchmarking, the true test of a high-resolution framework is its ability to accurately predict high-impact, localized weather events, where fine-scale physical details are critical. We therefore evaluated DeTI’s performance on two distinct and challenging types of extreme weather: tropical cyclones and regional heatwaves.

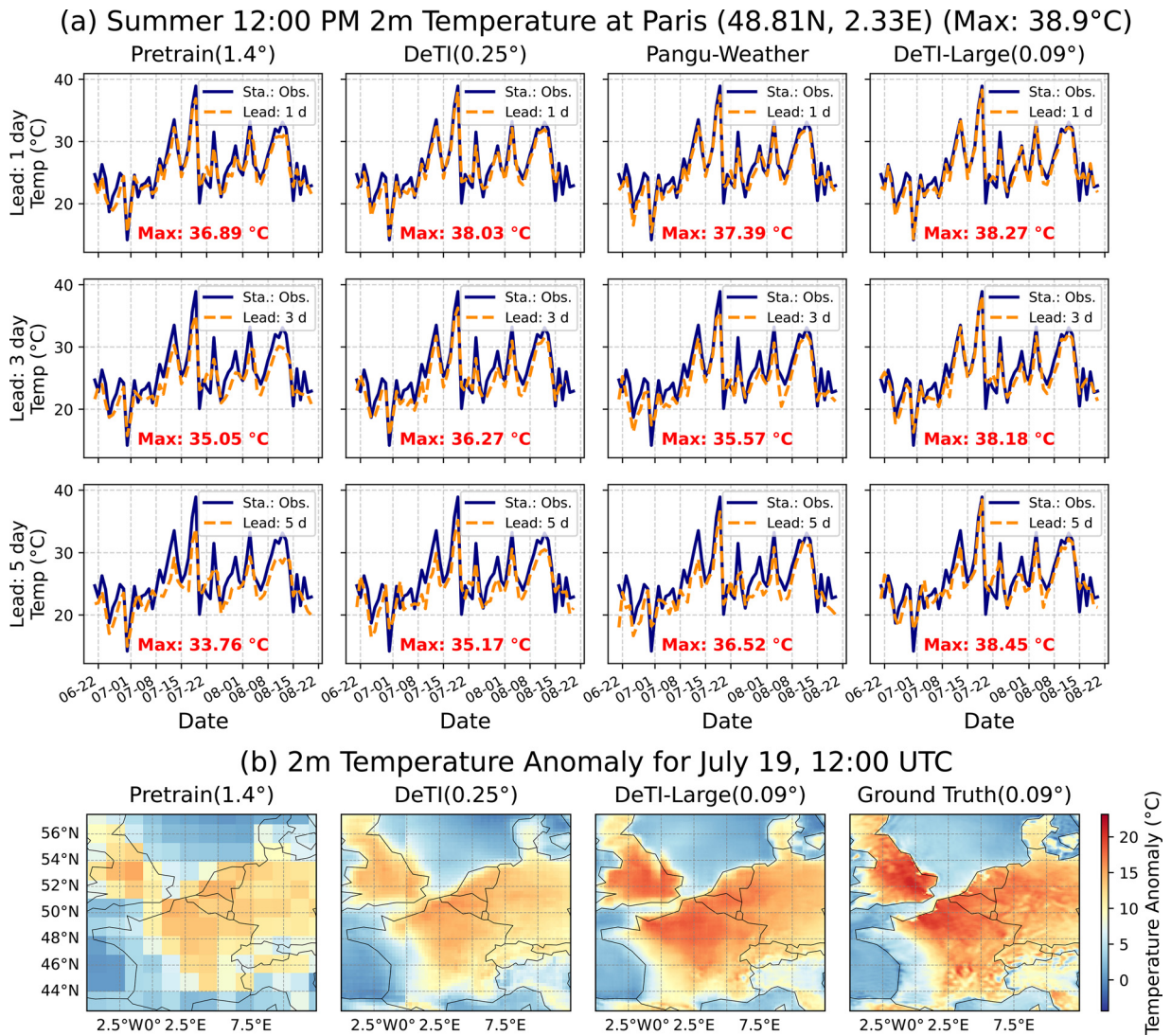
#### 3.2.1. Tropical cyclones

We first assess DeTI’s performance on 2022 tropical cyclones. We initialized all models with 2022 analysis data and used a standard detection algorithm, which is similar to that used in Aurora (Bodnar et al., 2024) (detailed in Text S11) to identify track position and intensity, which were then compared against real-world best-track data.

Fig. 3 presents the aggregate results, showing the mean direct position error and mean intensity error for all 2022 typhoons. We compare



**Fig. 3.** Comparison of typhoon forecasting performance: DeTI-Large (0.09°), DeTI (0.25°), Pangu-Weather, and GraphCast-Oper. (a) Mean direct position error (units: km) and (b) mean intensity error (units: m s<sup>-1</sup>) for 2022 typhoons as a function of forecast lead time (days). Visualizations of predicted tracks for (c) Typhoon Chaba and (d) Typhoon Hinnamnor. The “best” track (blue with circles), which is from the International Best Track Archive for Climate Stewardship (IBTrACS) (Gahtan et al., 2024), serves as the ground truth.



**Fig. 4.** Evaluation of 2m temperature (T2m) forecasts during the 2022 European summer heatwave. (a) Time series comparison of 1200 PM T2m at the Paris station (48.81°N, 2.33°E). Forecasts (orange dashed) from Pretrain (1.4°), DeTI (0.25°), Pangu-Weather, and DeTI-Large (0.09°) are compared against station observations (blue, solid) at 1-, 3-, and 5-day lead times. The maximum predicted temperature during the heatwave peak is annotated in red. (b) Spatial maps showing the 5-day forecast of the T2m anomaly for 1200 UTC 19 July 2022. Forecasts from Pretrain (1.4°), DeTI (0.25°), and DeTI-Large (0.09°) are compared to Ground Truth (0.09°).

our DeTI (0.25°) and DeTI-Large (0.09°) models against Pangu-Weather and GraphCast-Oper (which was fine-tuned on 2016–2021 data). As shown in Fig. 3(a), while DeTI (0.25°) exhibits a slightly larger track error, our high-resolution DeTI-Large (0.09°) model achieves a position error comparable to Pangu-Weather. We attribute the larger track error relative to GraphCast-Oper primarily to the fact that our models were not fine-tuned for this specific task, a step known from previous work to significantly improve track accuracy.

In contrast, Fig. 3(b) highlights a clear advantage for DeTI in intensity forecasting. Both DeTI models outperform the other baselines, and the high-resolution DeTI-Large (0.09°) model achieves the lowest mean intensity error overall. This strongly suggests that our framework's ability to resolve finer-scale atmospheric dynamics at higher resolutions directly translates to more accurate intensity predictions.

To further examine these statistics, we visualize the predicted tracks for two challenging case studies: Typhoon Chaba, which made a direct landfall, and Typhoon Hinnamor, which was characterized by a significant and sharp recursive turn. As shown in Fig. 3(c, d), DeTI's predicted paths are highly competitive. For both the landfall and the sharp turn, our models demonstrate comparable performance to the fine-tuned GraphCast-Oper and Pangu-Weather. This provides further evidence of

our framework's effectiveness and reliability in capturing the complex dynamics of extreme weather events, even without fine-tuning.

### 3.2.2. Heatwaves

Beyond the dynamic challenge of cyclones, we next assess DeTI's ability to capture high-impact thermodynamic extremes, where the fine-grained spatial detail of surface temperatures is critical. We focus on the 2022 European summer heatwave—an event characterized by record-breaking temperatures. First, Fig. 4(a) presents a time-series validation at a specific station in Paris (48.81°N, 2.33°E), comparing 1-, 3-, and 5-day forecasts against local station observations for 1200 PM 2-m temperature. While DeTI models achieve competitive RMSE scores, the ability to capture peak intensity is particularly notable. As annotated in the figure, for the 5-day forecast, DeTI-Large (0.09°) predicts a maximum temperature of 38.45°C, which is significantly closer to the observed peak and higher than the maximums predicted by Pangu-Weather (36.52°C) and the Pretrain model (33.76°C). This direct comparison of peak values demonstrates that the high-resolution framework effectively mitigates the smoothing effect often seen in lower-resolution forecasts, allowing for a more accurate representation of extreme heat magnitudes.

Moreover, Fig. 4(b) provides a spatial comparison of a 5-day forecast for the heatwave on 19 July 2022. To better visualize the event's severity and structure, we plot the temperature anomaly relative to the climatology and zoom in on western Europe. The low-resolution Pretrain (1.4°) model (left) produces a diffuse anomaly field, failing to capture the extreme intensity over the UK and France. In contrast, DeTI-Large (0.09°) (center-right) generates a sharp, high-fidelity anomaly map that closely mirrors the ground truth. It accurately reproduces the fine-grained spatial structure of the heat plume and the extreme positive anomalies (> 20°C anomaly), confirming that DeTI provides not only better global metrics but also more realistic and actionable regional forecasts for extreme thermodynamic events.

#### 4. Conclusions and future work

In this study, we present the DeTI framework, which removes a major computational barrier to training global high-resolution weather models. DeTI achieves a strong balance between forecast accuracy and efficiency, matching the performance of leading systems while using only a small fraction of their computational resources. For example, the 0.25° model was trained to full convergence within two days on eight GPUs. This efficiency comes from DeTI's two-part design, where a high-resolution regional model learns fine details under the guidance of a fixed, pretrained global model. Tests show that using this global guidance is essential for keeping forecasts stable and physically consistent. Furthermore, while our baseline configuration prioritizes efficiency, our analysis reveals that maximizing the patch size to the hardware limit can yield additional accuracy gains, albeit at the cost of increased training duration.

Beyond benchmark evaluations, DeTI also performs strongly in high-impact weather scenarios. It shows superior skill in predicting typhoon intensity and accurately captures the fine-scale spatial structure of heatwaves, demonstrating its potential for both operational forecasting and scientific research. These results confirm that the framework is not only computationally efficient but also capable of representing complex physical processes across scales.

Building on these findings, future work will focus on improving the interaction between the global and regional components to achieve stronger cross-scale coupling and more dynamic information exchange. Specifically, while the current GCI module mainly performs one-way context injections, future work will explore bidirectional information exchange to achieve stronger cross-scale coupling. Additionally, we intend to investigate importance-based sampling strategies, such as curriculum learning. By gradually shifting training focus from uniform sampling to regions with high meteorological variability or forecast error, we aim to further enhance the model's capability in handling extreme events. Another direction involves developing lightweight fine-tuning methods that can enhance variable-specific performance while maintaining efficiency. These efforts will help extend DeTI toward unified, kilometer-scale Earth system prediction. Together, these results establish DeTI as a practical and scalable foundation for next-generation weather and climate modeling, enabling high-resolution prediction without reliance on massive computing infrastructure.

#### Funding

This research was supported by the Smart-Grid National Science and Technology Major Project [grant number 2025ZD0805500] and Shanghai Municipal Commission of Economy and Informatization [grant number 2024-GZL-RGZN-02003].

#### Acknowledgements

The authors are grateful for the support provided by Shanghai Artificial Intelligence Laboratory.

#### Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.aosl.2026.100821.

#### References

- Alley, R.B., Emanuel, K.A., Zhang, F., 2019. Advances in weather prediction. *Science* 363 (6425), 342–344. doi:10.1126/science.aav7274.
- Ben-Bouallegue, Z., Clare, M.C., Magnusson, L., Gascon, E., Maier-Gerber, M., Janousek, M., Rodwell, M., et al., 2023. Rise data-driven weather forecast. *arXiv preprint. arXiv:2307.10128*.
- Bi, K., Xie, L., Zhang, H., Chen, X., Gu, X., Tian, Q., 2023. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619 (7970), 533–538. doi:10.1038/s41586-023-06185-3.
- Bodnar, C., Bruinsma, W.P., Lucic, A., Stanley, M., Brandstetter, J., Garvan, P., Riechert, M., et al., 2024. A foundation model for the Earth system. *arXiv preprint. arXiv:2405.13063*.
- Charney, J.G., Fjörtoft, R., Neumann, J.V., 1950. Numerical integration of the barotropic vorticity equation. *Tellus* 2 (4), 237–254. doi:10.3402/tellusa.v2i4.8607.
- Chen, K., Han, T., Gong, J., Bai, L., Ling, F., Luo, J.-J., Chen, X., et al., 2023a. Fengwu: Pushing the skillful global medium-range weather forecast beyond 10 days lead. *arXiv preprint. arXiv:2304.02948*.
- Chen, L., Zhong, X., Zhang, F., Cheng, Y., Xu, Y., Qi, Y., Li, H., 2023b. FuXi: A cascade machine learning forecasting system for 15-day global weather forecast. *Npj Clim. Atmos. Sci.* 6 (1), 190. doi:10.1038/s41612-023-00512-1.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., et al., 2020. An image is worth 16×16 words: Transformers for image recognition at scale. *arXiv preprint. arXiv:2010.11929v2*.
- Estevés, C., Slotine, J.-J., Makadia, A., 2023. Scaling spherical cnns. *arXiv preprint. arXiv:2306.05420*.
- Gahtan, J., Knapp, K.R., Schreck, C.J., Diamond, H.J., Kossin, J.P., Kruk, M.C., 2024. International best track archive for climate stewardship (IBTrACS) project, Version 4r01. NOAA National Centers for Environmental Information. doi:10.25921/82ty-9e16.
- Han, T., Guo, S., Ling, F., Chen, K., Gong, J., Luo, J., Gu, J., Dai, K., Ouyang, W., Bai, L., 2024. FengWu-GHR: Learning the kilometer-scale medium-range global weather forecasting. *arXiv preprint. arXiv:2402.00059*.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 27–30 June, Las Vegas, NV, USA, pp. 770–778. <https://cvpr2016.thecvf.com/>.
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horanyi, A., Munoz-Sabater, J., Nicolas, J., et al., 2020. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146 (730), 1999–2049. doi:10.24381/cds.adbb2d47.
- Hinton, G., Vinyals, O., Dean, J., 2015. Distilling the knowledge in a neural network. *arXiv preprint. arXiv:1503.02531*.
- Kochkov, D., Yuval, J., Langmore, I., Norgaard, P., Smith, J., Mooers, G., K'ower, M., et al., 2024. Neural general circulation models for weather and climate. *Nature* 632 (8027), 1060–1066. doi:10.1038/s41586-024-07744-y.
- Kurth, T., Subramanian, S., Harrington, P., Pathak, J., Mardani, M., Hall, D., Miele, A., Kashinath, K., Anandkumar, A., 2023. FourCastNet: Accelerating global high-resolution weather forecasting using adaptive fourier neural operators. In: *Proceedings of the Platform for Advanced Scientific Computing Conference (PASC '23)* 26–28 June, Davos, Switzerland. <https://pasc23.pasc-conference.org/>.
- Lam, R., Sanchez-Gonzalez, A., Willson, M., Wirsberger, P., Fortunato, M., Pritzel, A., Ravuri, S., et al., 2022. Learning skillful medium-range global weather forecasting. *arXiv preprint. arXiv:2212.12794*.
- Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B., 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 11–17 October, Virtual Event, pp. 10012–10022. <https://iccv2021.thecvf.com/>.
- Micikevicius, P., Narang, S., Alben, J., Diamos, G., Elsen, E., Garcia, D., Ginsburg, B., et al., 2018. Mixed precision training. *arXiv preprint. arXiv:1710.03740*.
- Price, I., Sanchez-Gonzalez, A., Alet, F., Andersson, T.R., El-Kadi, A., Masters, D., Ewalds, T., et al., 2023. GenCast: Diffusion-based ensemble forecasting for medium-range weather. *arXiv preprint. arXiv:2312.15796*.
- Ramavajjala, V., 2024. HEAL-ViT: Vision transformers on a spherical mesh for medium-range weather forecasting. *arXiv preprint. arXiv:2403.17016*.
- Rasp, S., Dueben, P.D., Scher, S., Weyn, J.A., Mouatadid, S., Thuerey, N., 2020. WeatherBench: A benchmark data set for data-driven weather forecasting. *J. Adv. Model. Earth Syst.* 12 (11), e2020MS002203. doi:10.1029/2020MS002203.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation. In: *Proc. 18th International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015)*, October 5–9, Munich, Germany, pp. 234–241. <https://miccai2015.org/>.
- Schmidt, R.M., 2019. Recurrent Neural Networks (RNNs): A gentle introduction and overview. *arXiv preprint. arXiv:1912.05911*.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2023. Attention Is all you need. *arXiv preprint. arXiv:1706.03762*.
- Zhao, Y., Gu, A., Varma, R., Luo, L., Huang, C.-C., Xu, M., Wright, L., et al., 2023. PyTorch FSDP: Experiences on scaling fully sharded data parallel. *arXiv preprint. arXiv:2304.11277*.
- Zhong, X., Chen, L., Fan, X., Qian, W., Liu, J., Li, H., 2024. FuXi-2.0: An advanced machine learning weather forecasting model for practical applications. *arXiv preprint. arXiv:2409.07188*.